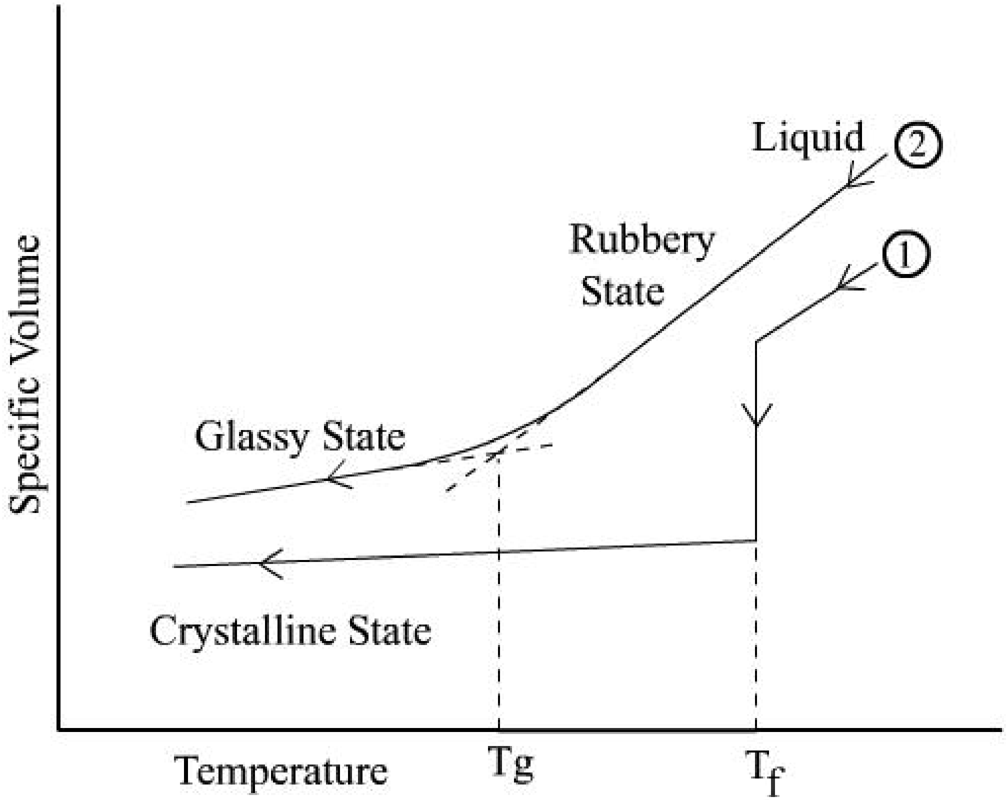
How to succeed

- Read the Chapter ahead of lectures
- Come to class
- **Start paper early**
- Study groups
- Prepare exams
- Don’t cheat, plagiarize, or otherwise participate in un-ethical behavior
- **Learn with Internet: Wikipedia, google(baidu)…**
- Ask questions and Think skeptically

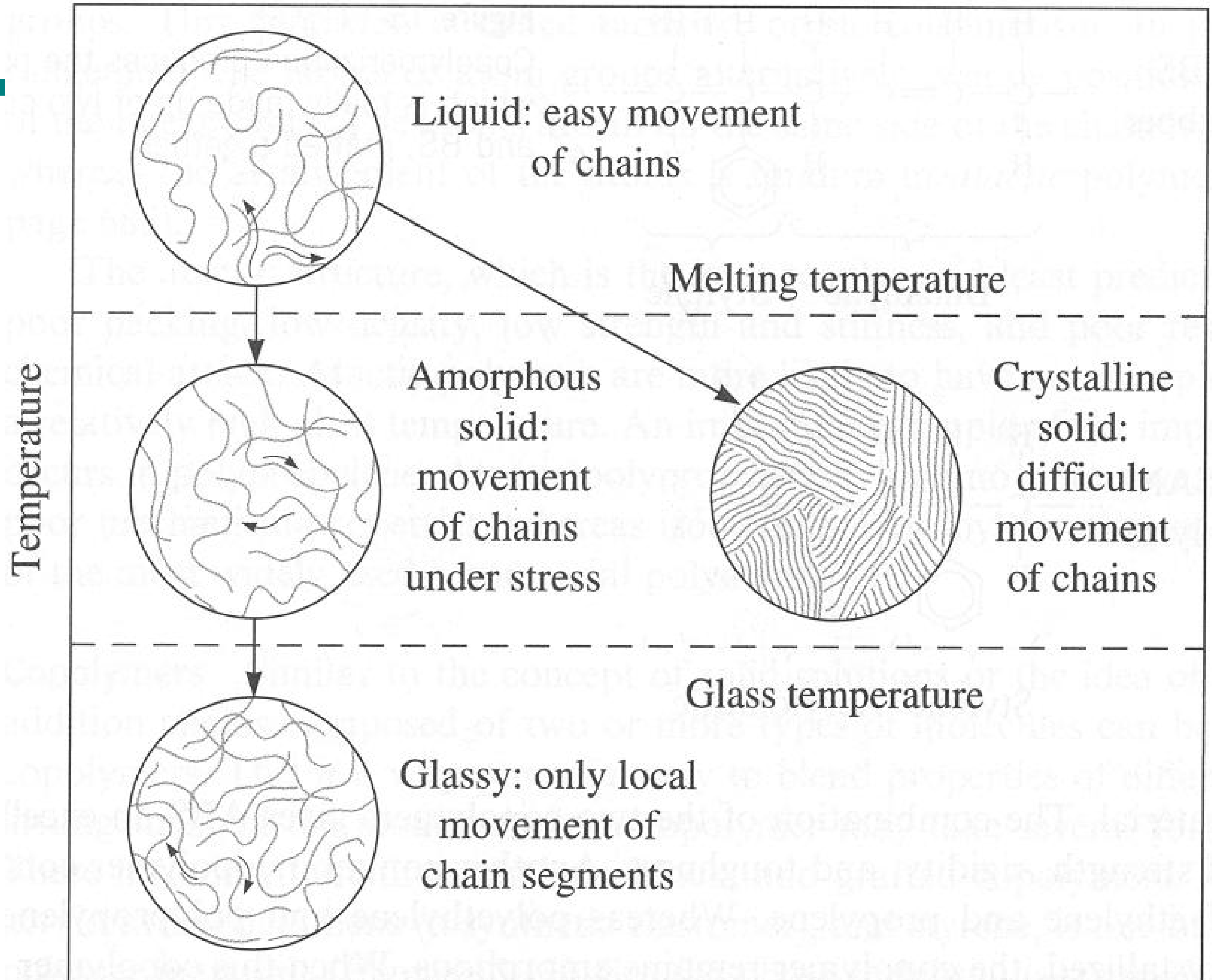
Important Points from Last Lecture

- Polymer Crystals
- Melting of Polymers



Lecture 4-5:
Solid State Properties III

SUN Dazhi


Liquid: easy movement of chains
Melting temperature
Amorphous solid: movement of chains under stress
Crystalline solid: difficult movement of chains
Glass temperature
Glassy: only local movement of chain segments
Temperature

Polymer Phases

- **Viscous Liquid**

Polydimethylsiloxane T_g = -123° C; T_m = -40 ° C

- **Elastomeric**

Polyisoprene T_g = -73 ^oC

Polybutadiene, T_g = -85 ^oC

Polychloroprene, T_g = -50 ^oC

Polyisobutylene, T_g = -70 ^oC

- **Amorphous Glassy**

Polystyrene T_g = 100 ^oC

Polymethyl methacrylate, T_g = 105 ^oC

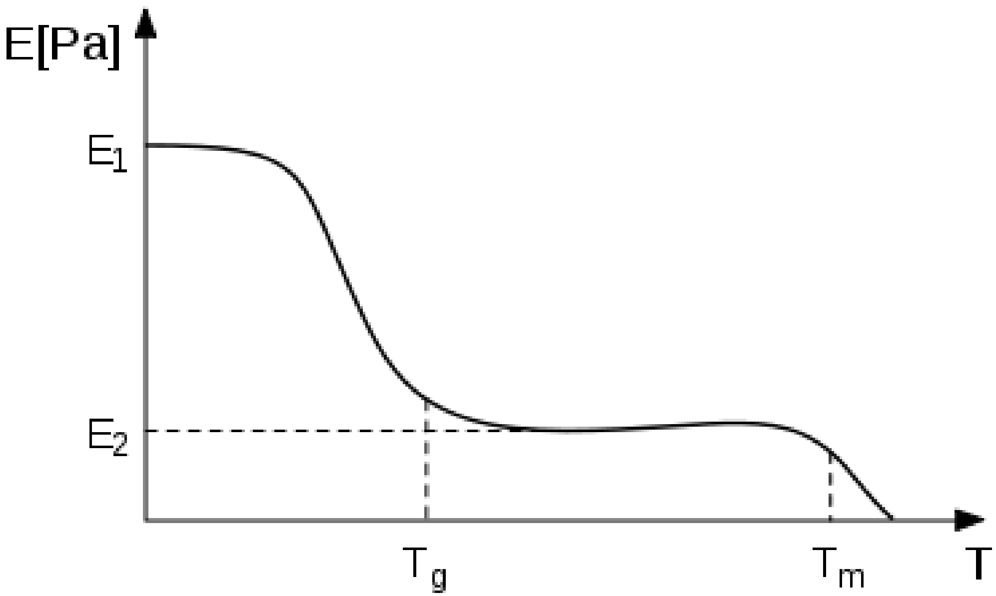
- ***Semi*-Crystalline**

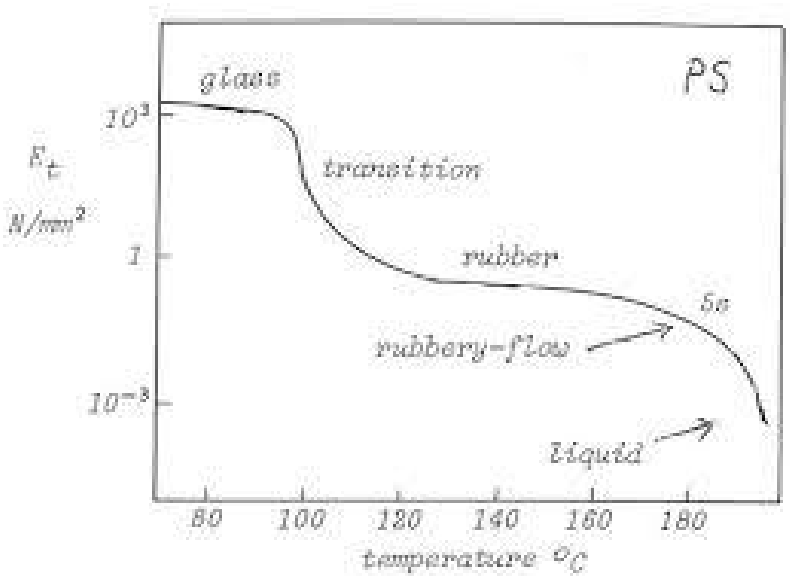
Nylon 6,6, T_g = 50 ^oC; T_m = 265 ^oC

PET, T_g = 65 ^oC; T_m =270 ^oC

Glass-Rubber-Liquid

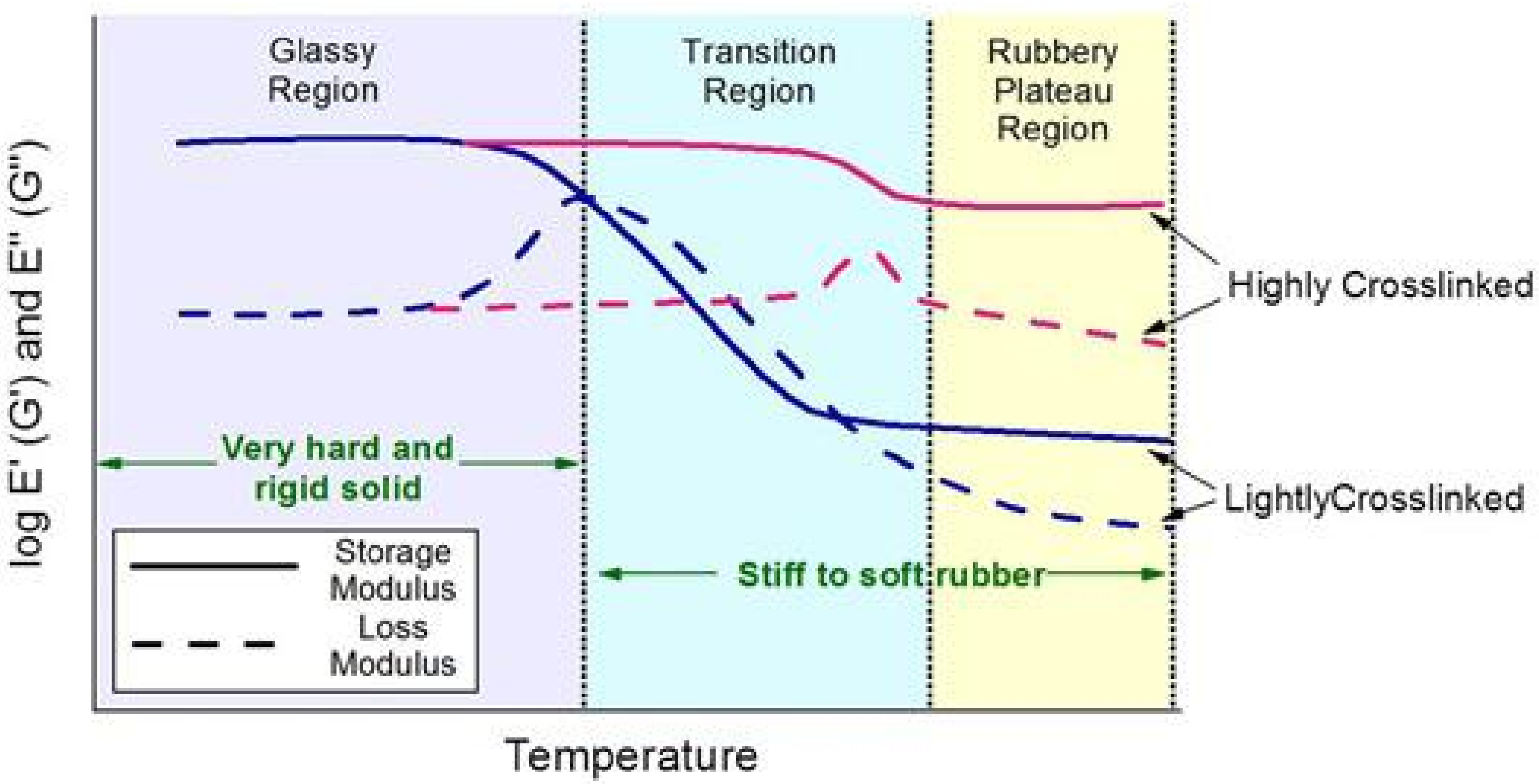
- **Amorphous plastics have a complex thermal profile with 3 typical states:**



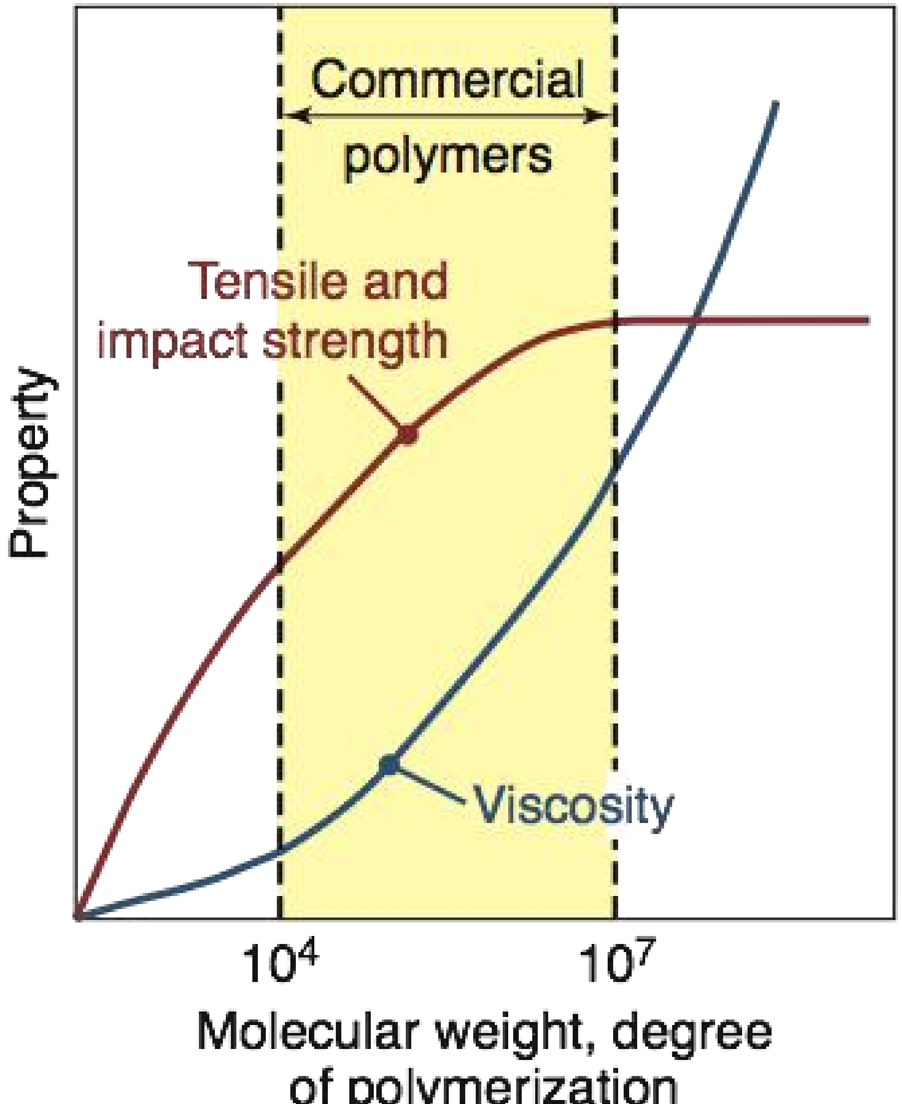


Glass-Rubber-Liquid

- **Crosslinked polymers**

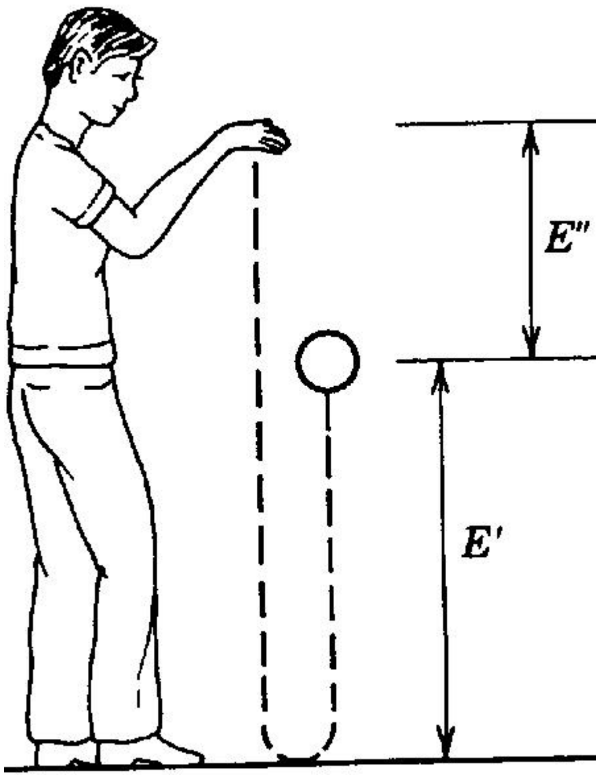


Mechanical Properties



Transient Testing

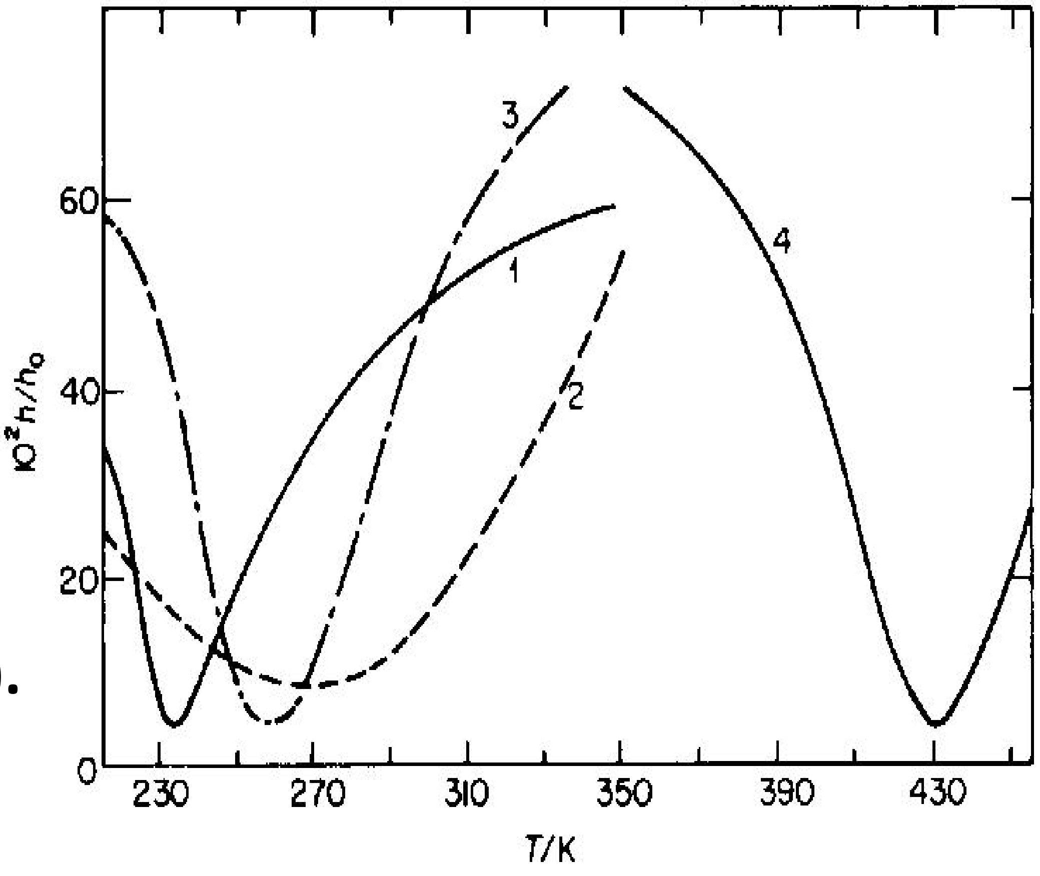
Resilience of Cured Elastomers



- Resilience tests reflect the ability of an elastomeric compound to store and return energy at a given frequency and temperature.

Resilience of Cured Elastomers

Change of rebound resilience (h/ho) with temperature T for:

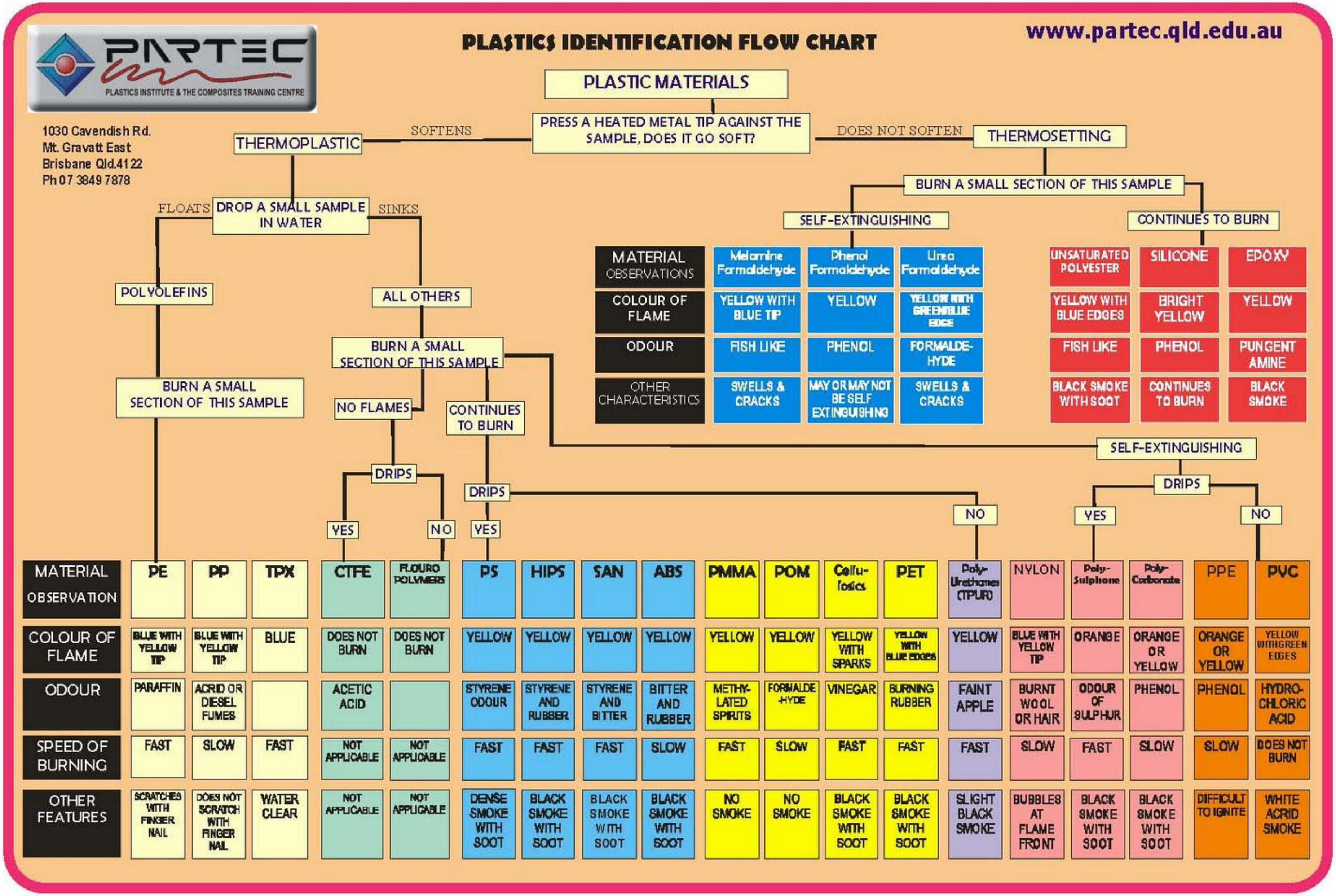


1. *cis*-poly(isoprene);
2. poly(isobutylene);
3. poly(chloroprene);
4. poly(methyl methacrylate).

Types of Polymers

Polymer Family Tree

- **Thermoplastics**
  Will reform when melted
  - **Not Cross-Linked**
    90% of market
    - Polyethylene
      33%
    - Vinyls
      16%
    - Polypropylene
      15%
    - PMMA
    - ABS
    - Nylon
    - Polycarbonate
    - Saturated Polyester
    - PEEK
    - Polyurethane
      Some are thermosets as well.
    - PVC
- **Thermosets/Elastomers**
  Will not reform
  - **Cross-linked**
    10% of market
    - Epoxy
    - Melamine Formaldehyde
    - Phenolic
    - Polyester (unsaturated)
    - Polyimide
    - Polyurethane
      Some are thermoplastic as well.
    - Silicone
    - Urea Formaldehyde



MATERIAL OBSERVATIONS	Melamine Formaldehyde	Phenol Formaldehyde	Urea Formaldehyde	UNSATURATED POLYESTER	SILICONE	EPOXY
COLOUR OF FLAME	YELLOW WITH BLUE TIP	YELLOW	YELLOW WITH GREEN/BLUE EDGE	YELLOW WITH BLUE EDGES	BRIGHT YELLOW	YELLOW
ODOUR	FISH LIKE	PHENOL	FORMALDE-HYDE	FISH LIKE	PHENOL	PUNGENT AMINE
OTHER CHARACTERISTICS	SWELLS & CRACKS	MAY OR MAY NOT BE SELF EXTINGUISHING	SWELLS & CRACKS	BLACK SMOKE WITH SOOT	CONTINUES TO BURN	BLACK SMOKE

MATERIAL OBSERVATION	PE	PP	TPX	CTFE	FLOURO POLYMERS	PS	HIPS	SAN	ABS	PMMA	POM	Cellu-losics	PET	Poly-Urethane (TPUR)	NYLON	Poly-Sulphone	Poly-Carbonate	PPE	PVC
COLOUR OF FLAME	BLUE WITH YELLOW TIP	BLUE WITH YELLOW TIP	BLUE	DOES NOT BURN	DOES NOT BURN	YELLOW	YELLOW	YELLOW	YELLOW	YELLOW	YELLOW	YELLOW WITH SPARKS	YELLOW WITH BLUE EDGES	YELLOW	BLUE WITH YELLOW TIP	ORANGE	ORANGE OR YELLOW	ORANGE OR YELLOW	YELLOW WITH GREEN EDGES
ODOUR	PARAFFIN	ACRID OR DIESEL FUMES		ACETIC ACID		STYRENE ODOUR	STYRENE AND RUBBER	STYRENE AND BITTER	BITTER AND RUBBER	METHYLATED SPIRITS	FORMALDE-HYDE	VINEGAR	BURNING RUBBER	FAINT APPLE	BURNT WOOL OR HAIR	ODOUR OF SULPHUR	PHENOL	PHENOL	HYDRO-CHLORIC ACID
SPEED OF BURNING	FAST	SLOW	FAST	NOT APPLICABLE	NOT APPLICABLE	FAST	FAST	FAST	SLOW	FAST	SLOW	FAST	FAST	FAST	SLOW	FAST	SLOW	SLOW	DOES NOT BURN
OTHER FEATURES	SCRATCHES WITH FINGER NAIL	DOES NOT SCRATCH WITH FINGER NAIL	WATER CLEAR	NOT APPLICABLE	NOT APPLICABLE	DENSE SMOKE WITH SOOT	BLACK SMOKE WITH SOOT	BLACK SMOKE WITH SOOT	BLACK SMOKE WITH SOOT	NO SMOKE	NO SMOKE	BLACK SMOKE WITH SOOT	BLACK SMOKE WITH SOOT	SLIGHT BLACK SMOKE	BUBBLES AT FLAME FRONT	BLACK SMOKE WITH SOOT	BLACK SMOKE WITH SOOT	DIFFICULT TO IGNITE	WHITE ACRID SMOKE

Ballpark Comparisons

- **Tensile strengths**

  Polymers: ~ 10 - 100 MPa

  Metals: 100's - 1000's MPa

- **Elongation**

  Polymers: up to 1000 % in some cases

  Metals: < 100%

- **Moduli (Elastic or Young's)**

  Polymers: ~ 10 MPa - 4 GPa

  Metals: ~ 50 - 400 GPa

Stress Strain Studies

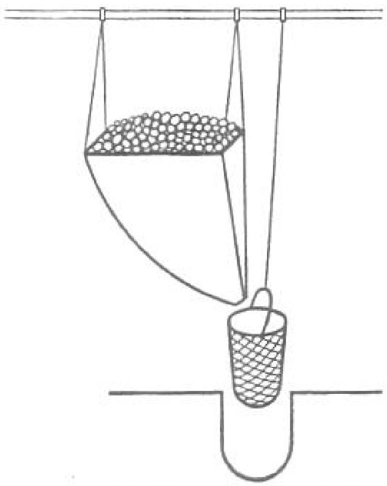


FIG. 10. Tensile test of wire by Leonardo da Vinci.

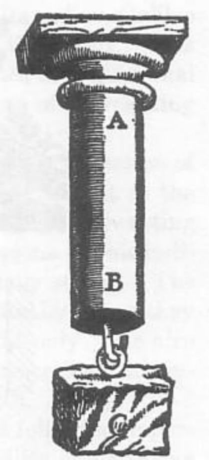
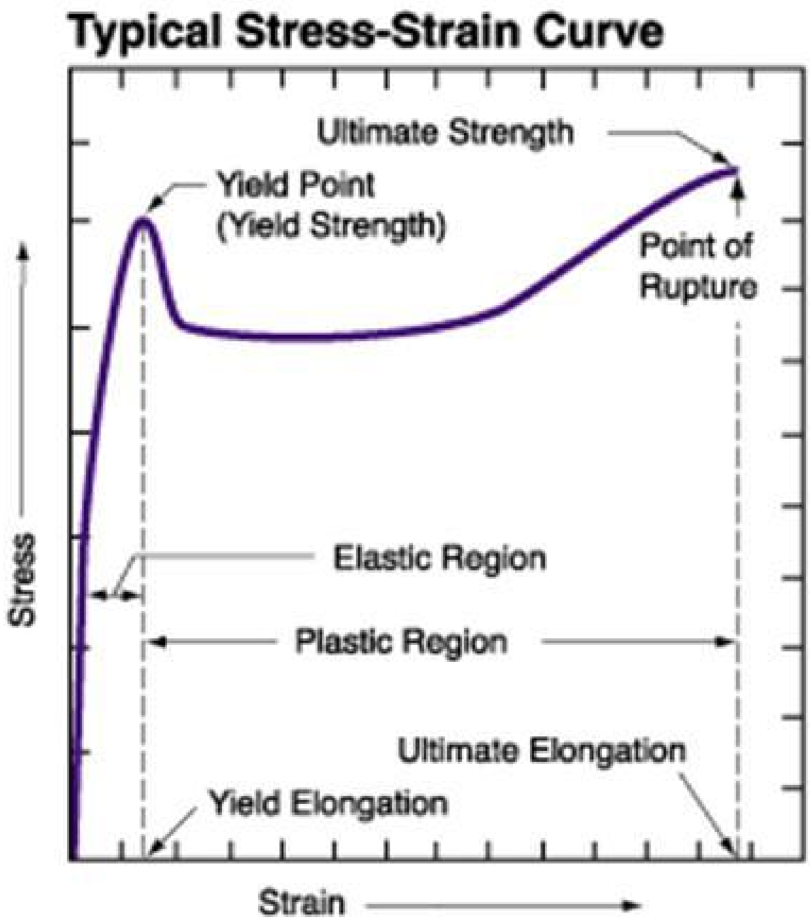


FIG. 14. Galileo's illustration of tensile test.

Stress-Strain Curve



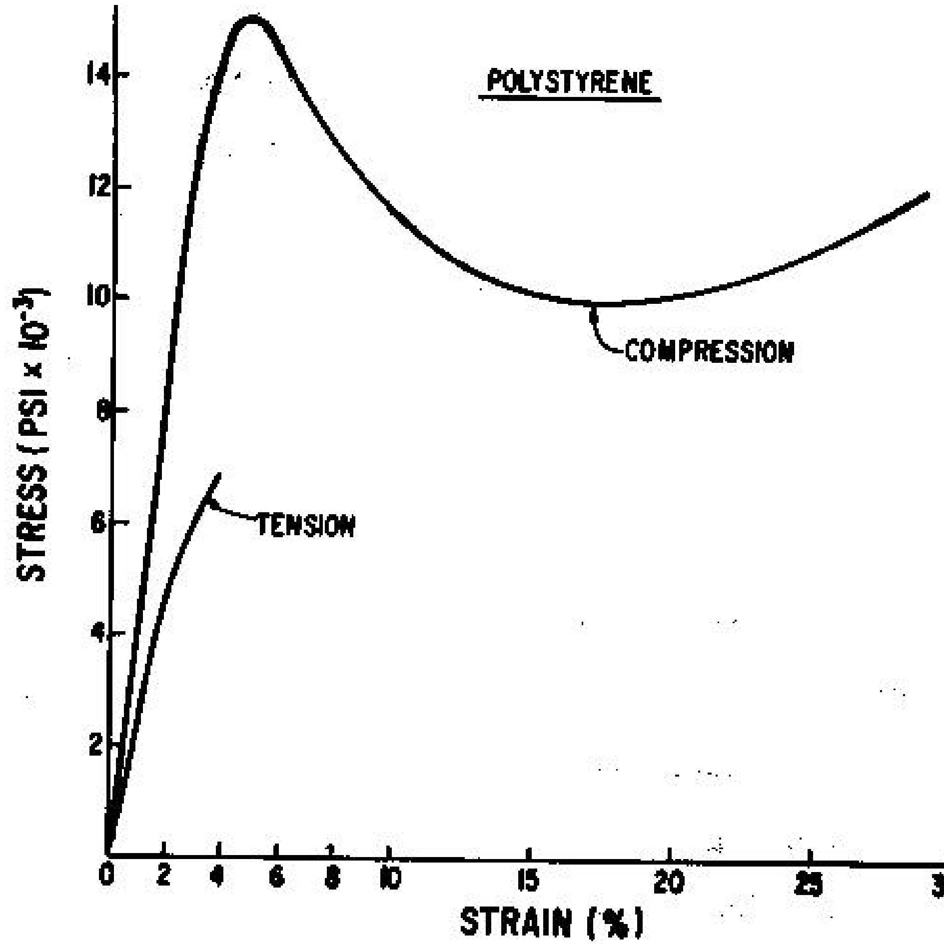
- Initial slope is the Young's Modulus (E' or sometimes G)
- TS = tensile strength
- σ_y = yield strength
- Toughness = Energy required to break (area under curve)

Compression vs. Tensile Tests

Stress-strain curves are very dependent on the test method

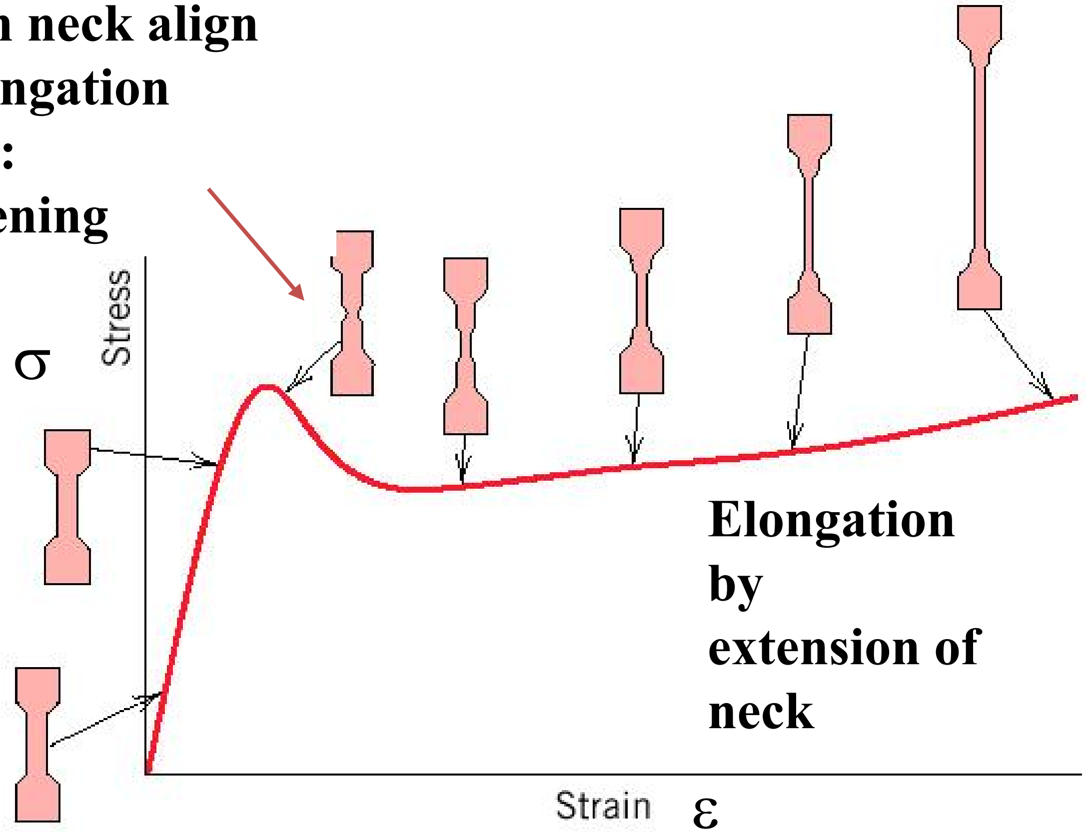
- Tensile testing is most sensitive to material flaws and microscopic cracks.

- Compression tests tend to be characteristic of the polymer, while tension tests are more characteristic of sample flaws.



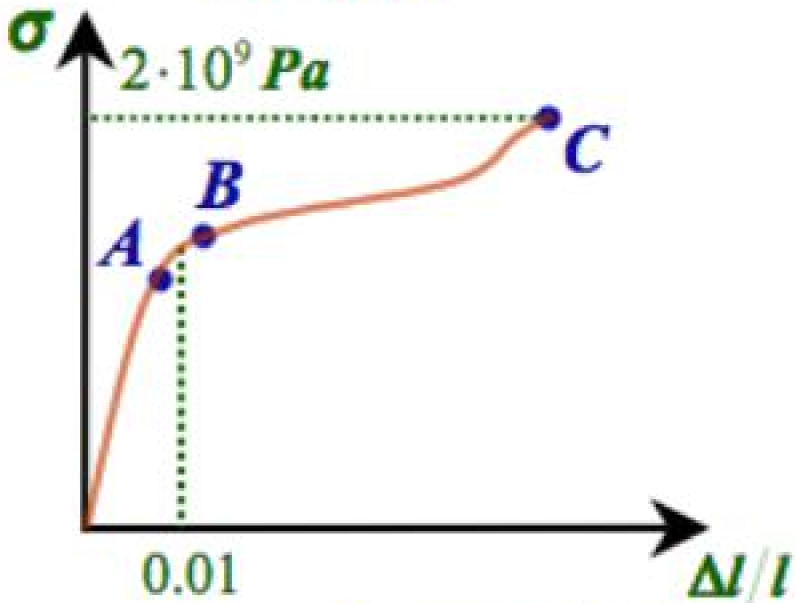
Stress Strain Graphs

Chains in neck align along elongation direction: strengthening

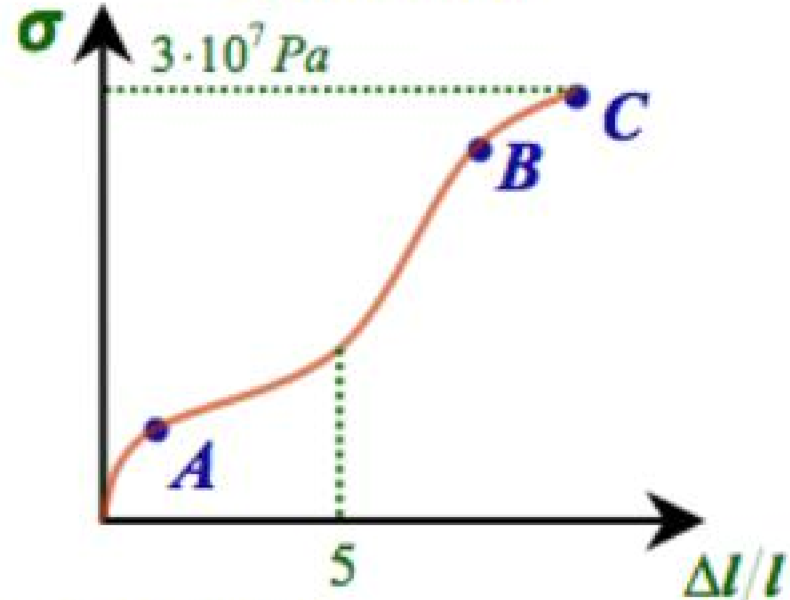


Elongation by extension of neck

For steel



For rubber

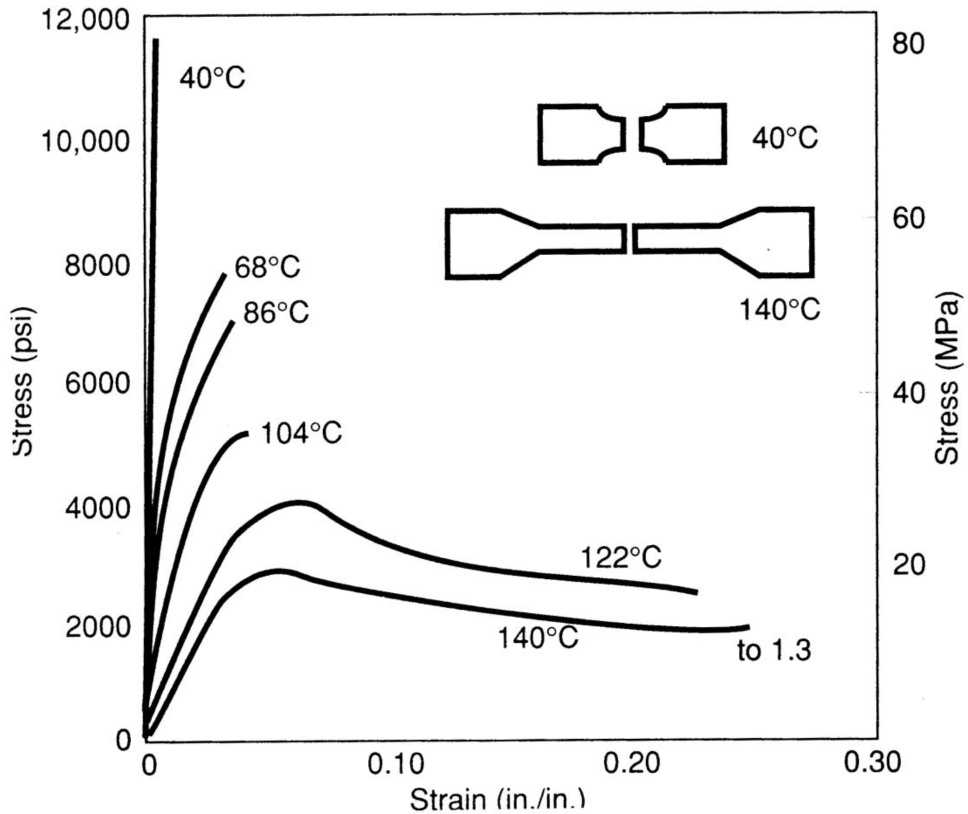


A - upper limit for stress-strain linearity

B - upper limit for reversibility of deformations

C - fracture point

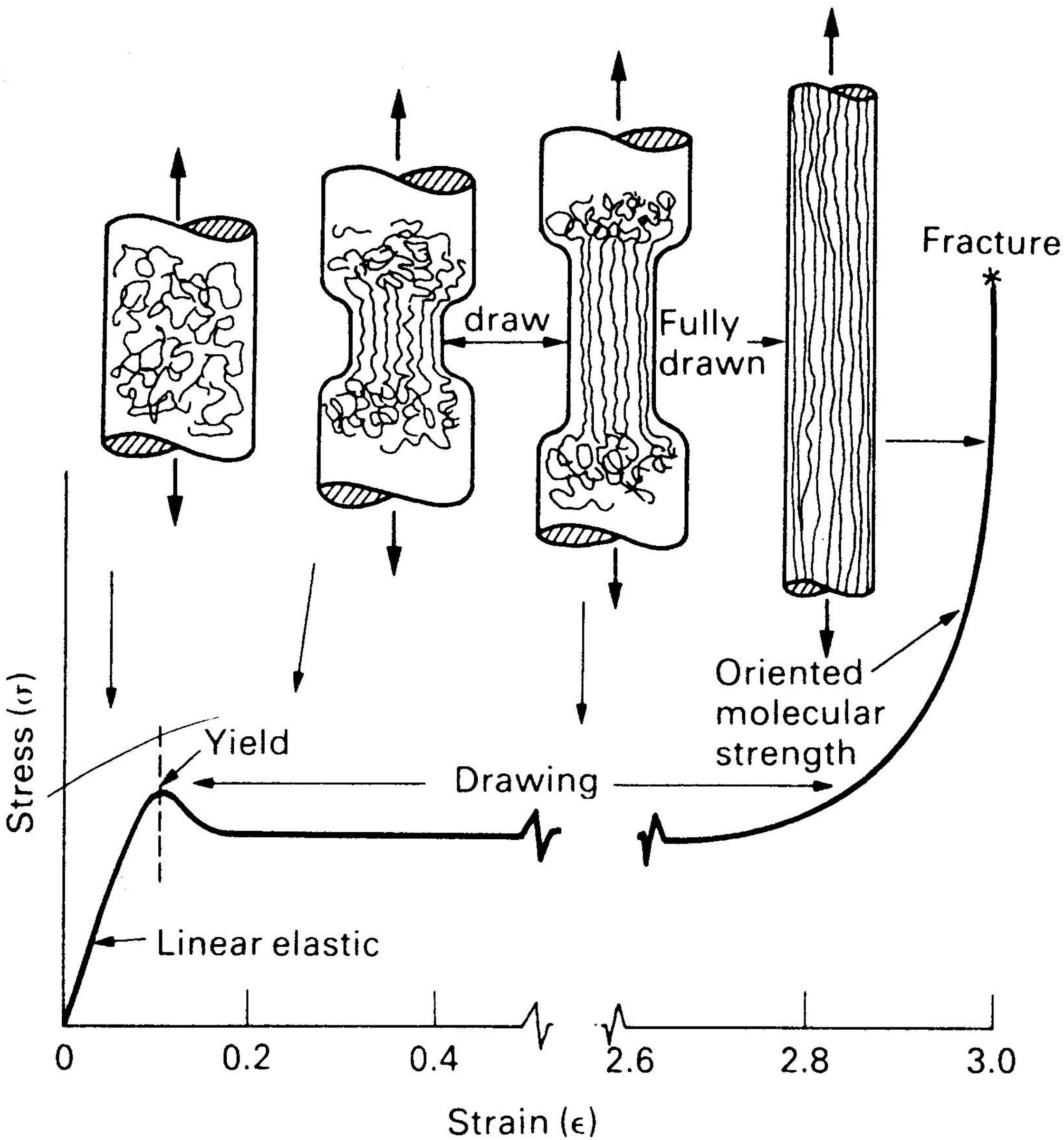
Ductility & Elongation (E_L)



E_L < 5% Brittle
E_L > 5% Ductile

Thermosets = strong & brittle
Not Ductile

Thermolastics = depends on T

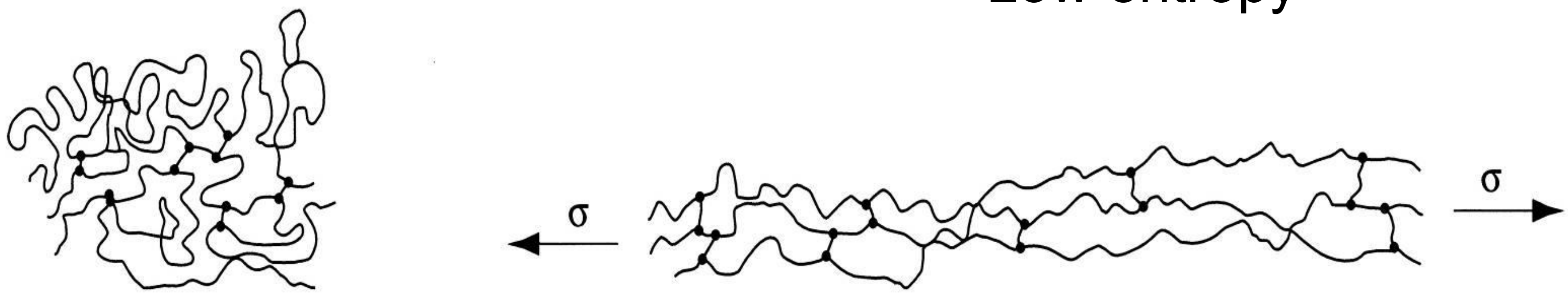


Fracture
draw
Fully drawn
Oriented molecular strength
Yield
Drawing
Linear elastic
Stress (σ)
0
0.2
0.4
2.6
2.8
3.0
Strain (ϵ)

Elastomer Molecules

High entropy

Low entropy

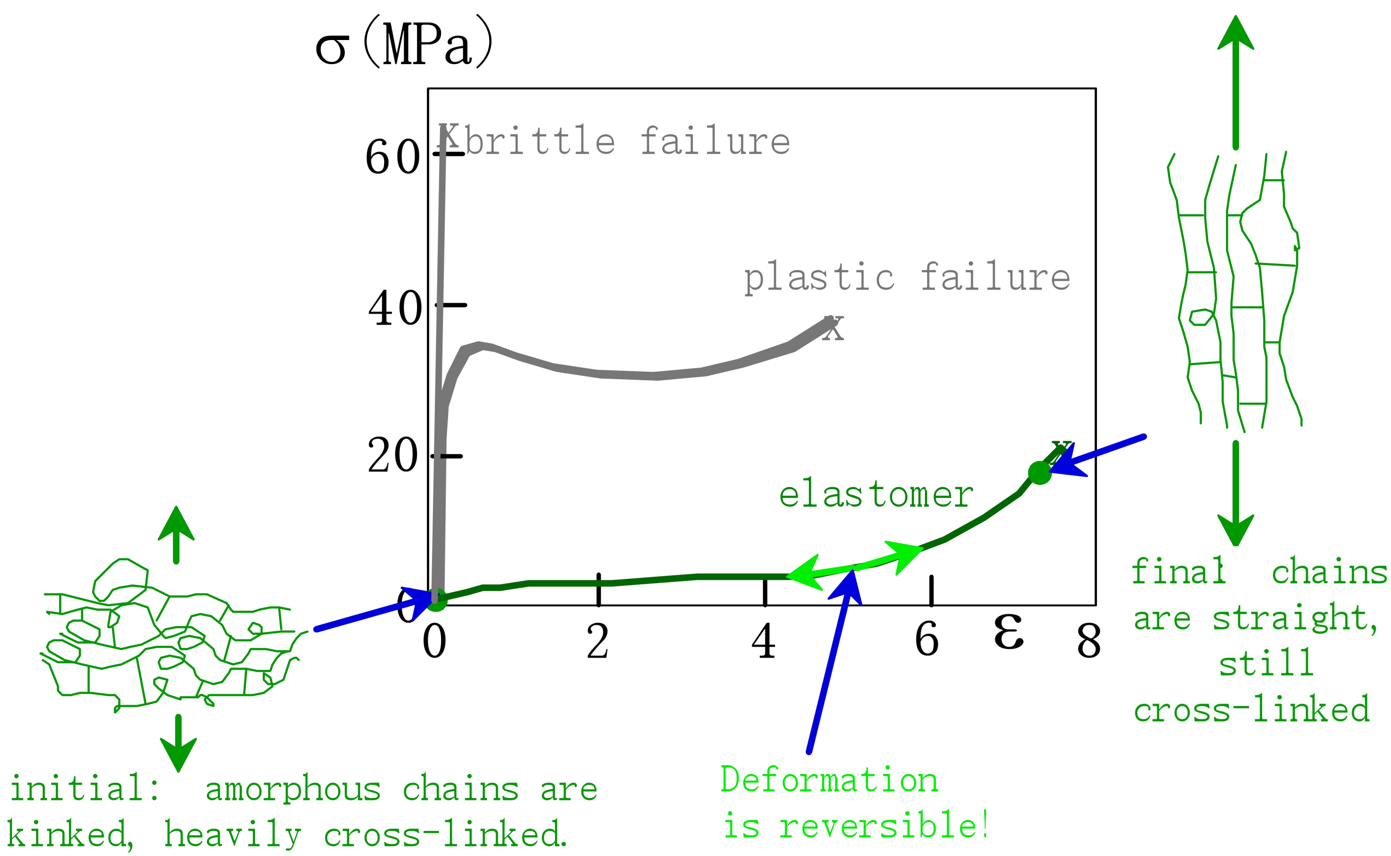


(a) (b)

Low energy

High energy

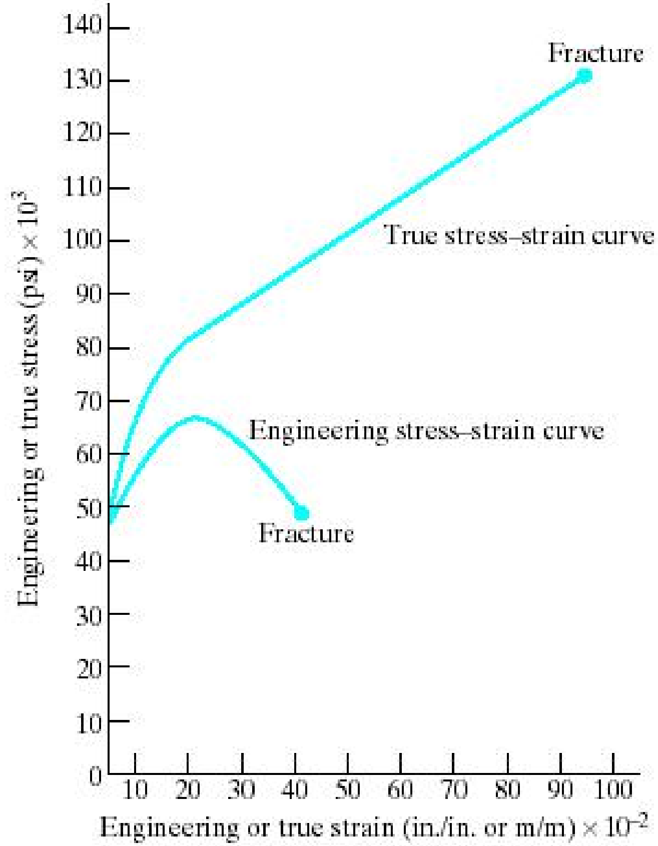
Elastomer: TENSILE RESPONSE



- Compare to responses of other polymers:
  --brittle response (aligned, cross linked & networked case)
  --plastic response (semi-crystalline case)

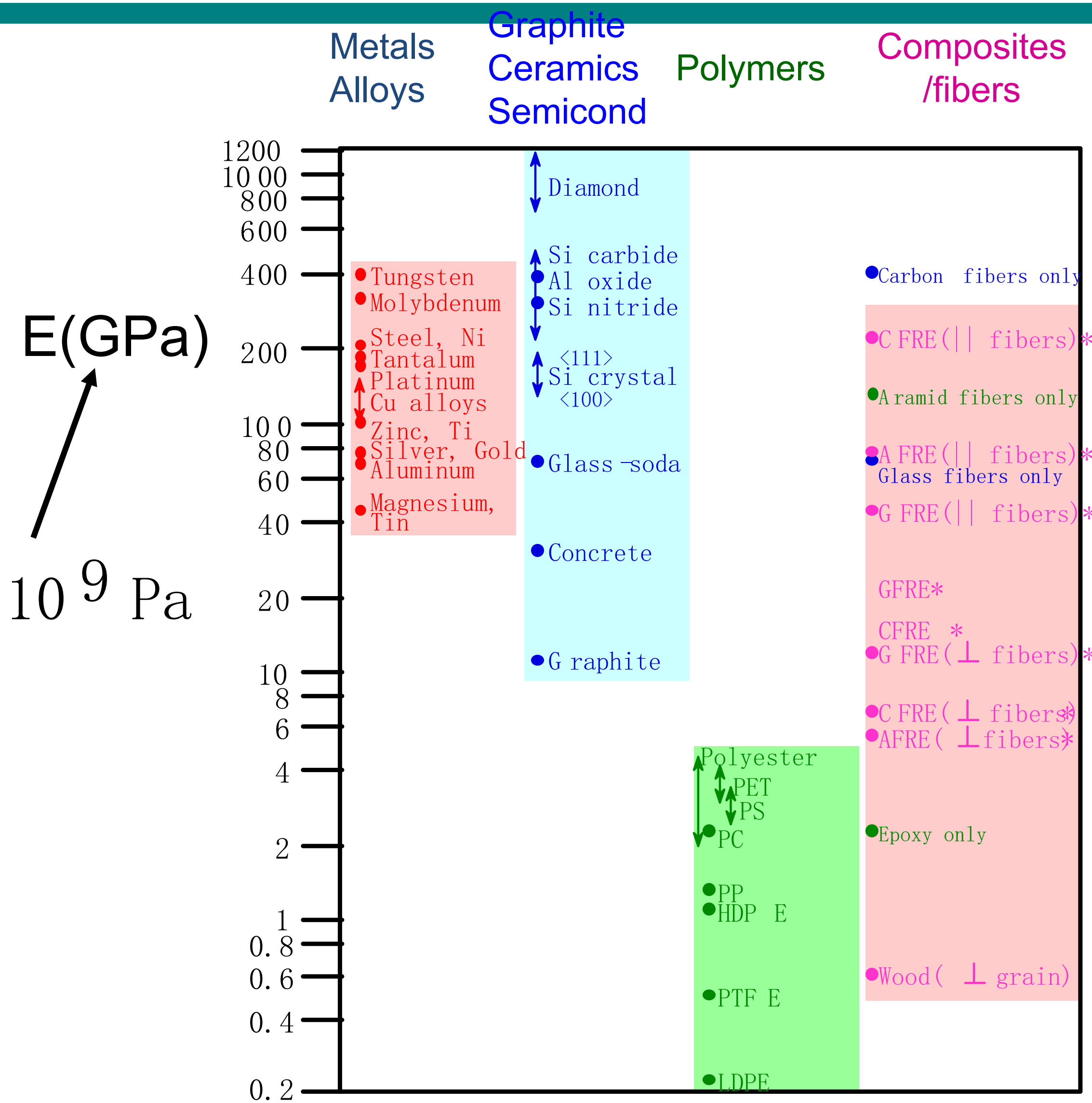
True Stress-strain Curve

Figure 6-7 *Neck down of a tensile test specimen within its gage length after extension beyond the tensile strength. (Courtesy of R. S. Wortman)*



igure 6-8 *True stress (= load divided by actual area in the necked-down region) continues to rise to the point of fracture, in contrast to the behavior of engineering stress. (After R. A. Flinn/P. K. Trojan:* Engineering Materials and Their Applications, *2nd Ed., Copyright © 1981, Houghton Mifflin Company, used by permission.)*

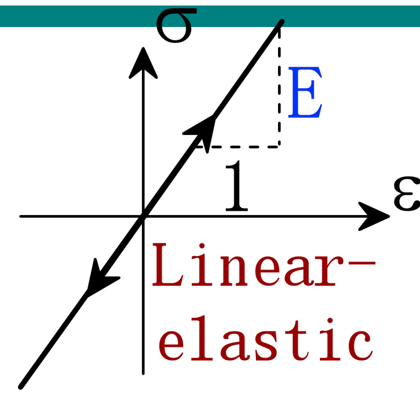
YOUNG'S MODULI: COMPARISON



Based on data in Table B2, *Callister 6e*.
Composite data based on reinforced epoxy with 60 vol% of aligned carbon (CFRE), aramid (AFRE), or glass (GFRE) fibers.

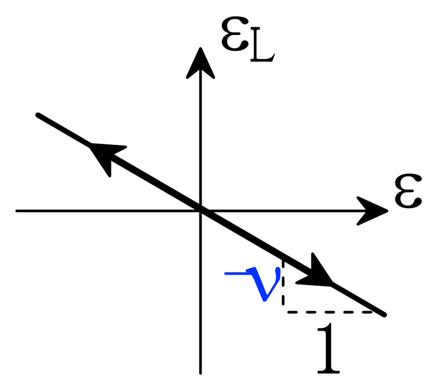
Linear Elasticity: Possion Effect

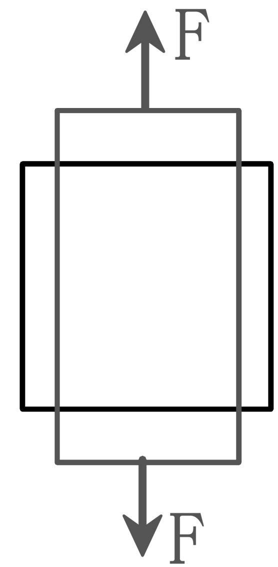
- **Hooke's Law: $\sigma = E\,\varepsilon$**



- **Poisson's ratio, ν:**

$$\nu = -\frac{\textit{width strain}}{\textit{axial strain}} = -\frac{\Delta w / w}{\Delta l / l} = -\frac{\varepsilon_L}{\varepsilon}$$





simple
tension
test

metals: ~ 0.33
ceramics: ~0.25
polymers: ~0.40

Units:
E: [GPa] or [psi]
ν: dimensionless

Why does ν have minus sign?

Poisson Ratio: materials specific

Metals:	Ir	W	Ni	Cu	Al	Ag	Au
	0.26	0.29	0.31	0.34	0.34	0.38	0.42

generic value ~ 1/3

Solid Argon: 0.25

Covalent Solids:	Si	Ge	Al_2O_3	TiC
	0.27	0.28	0.23	0.19

generic value ~ 1/4

Ionic Solids: MgO 0.19

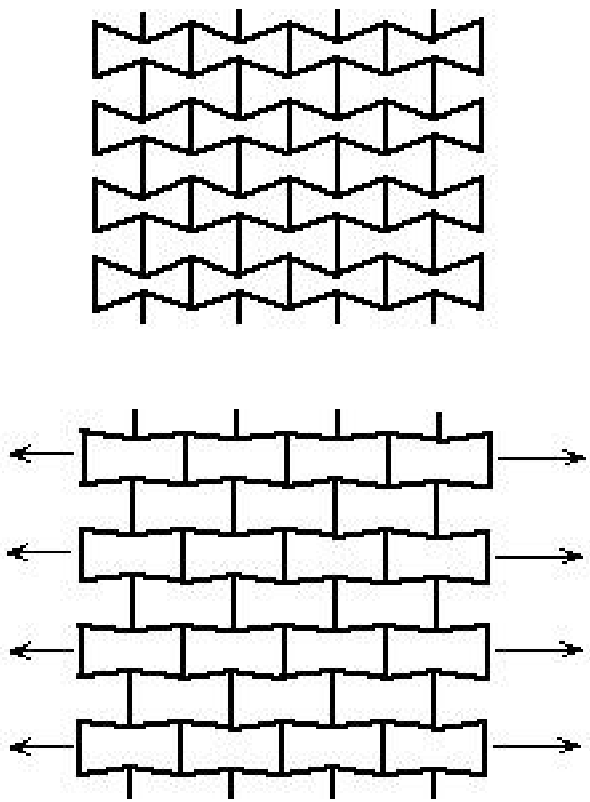
Silica Glass: 0.20

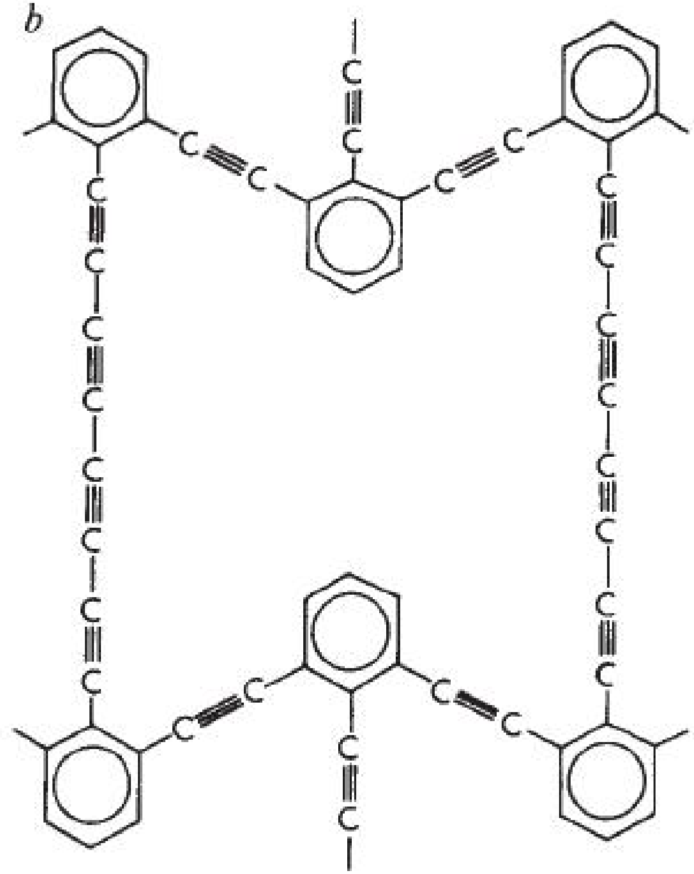
Polymers: Network (Bakelite) 0.49 Chain (PE) 0.40

Elastomer: Hard Rubber (Ebonite) 0.39 (Natural) 0.49

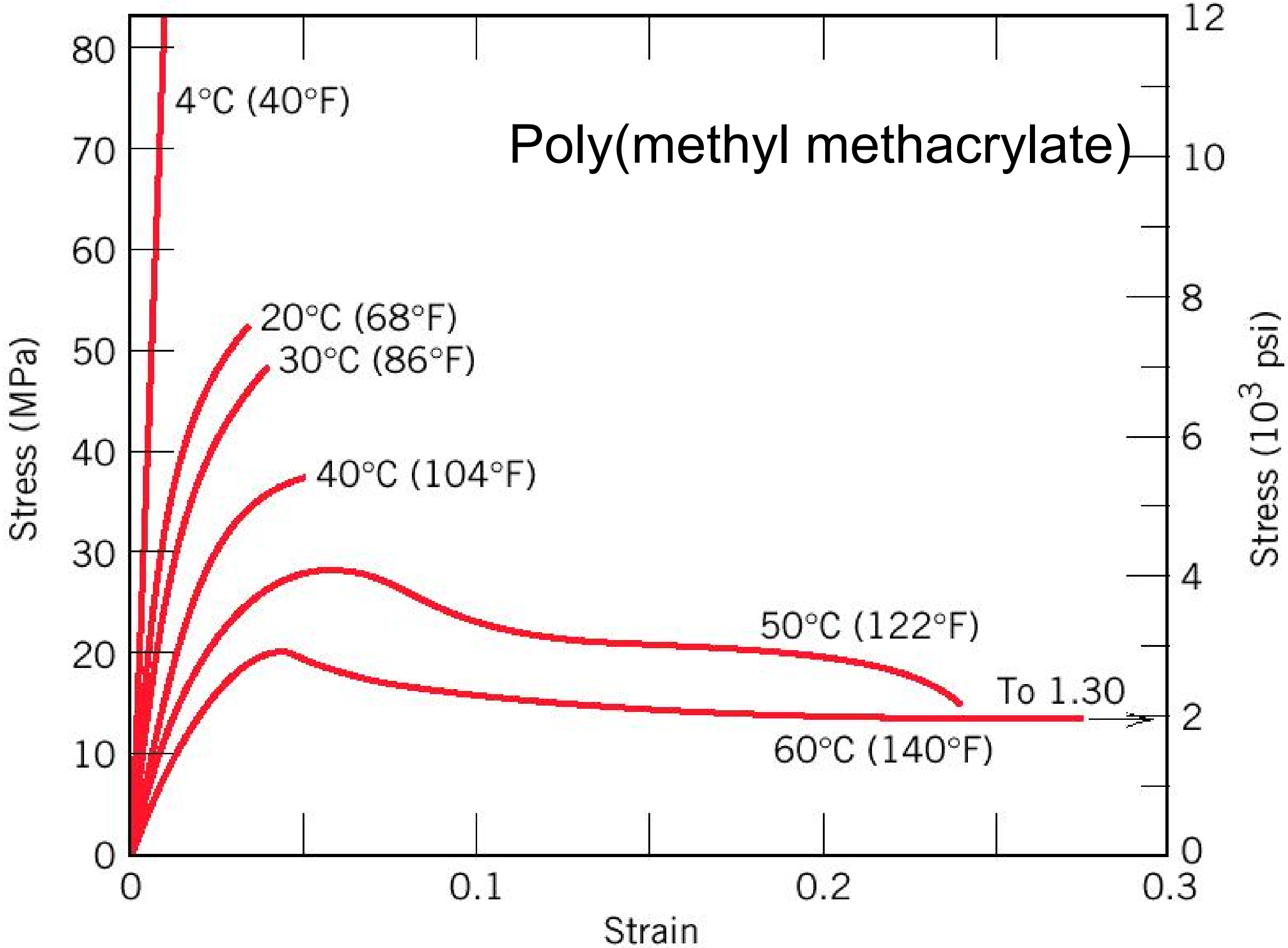
Negtive poisson’s ratio

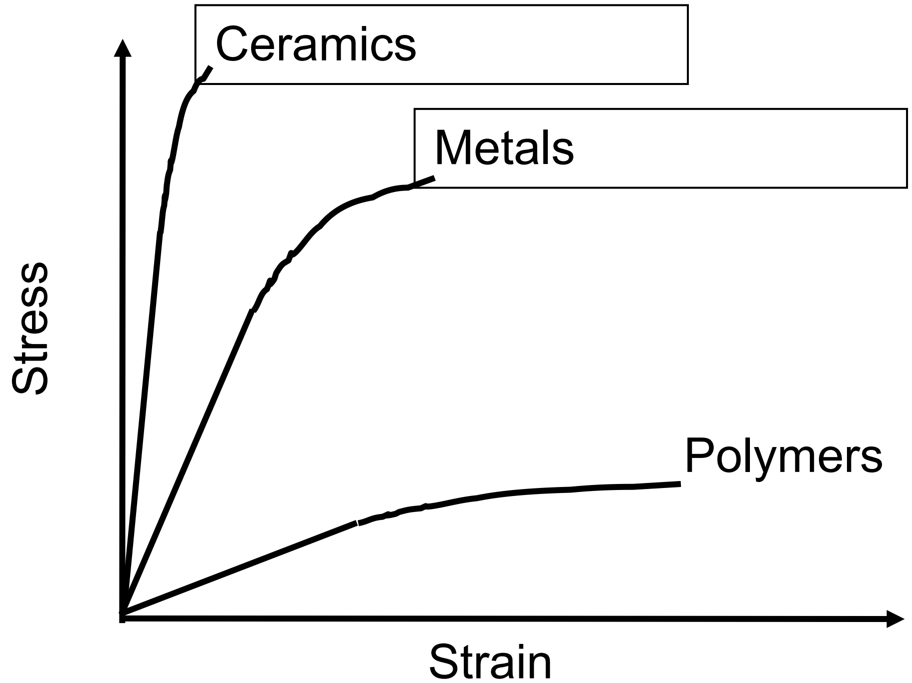
- foams



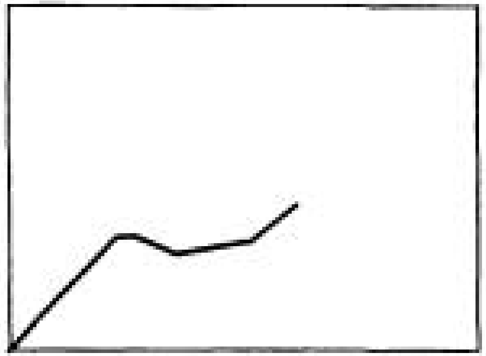


Mechanical properties are sensitive to temperature

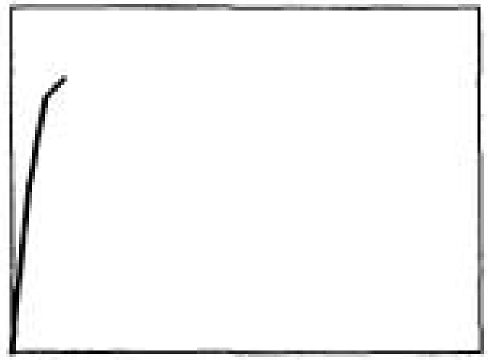




- Lower elastic modulus, yield and ultimate properties
- Greater post-yield deformability
- Greater failure strain

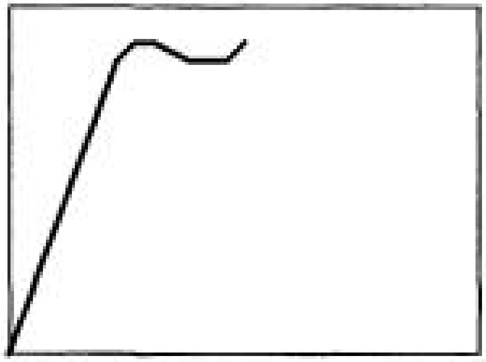


Soft and weak

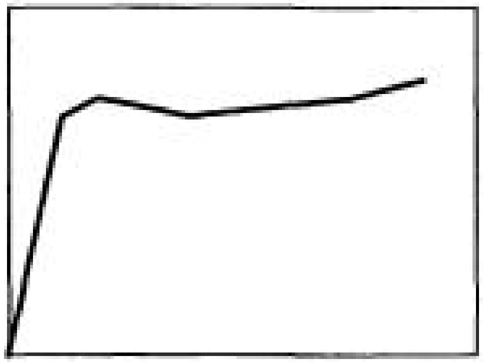


Hard and brittle

Soft and tough

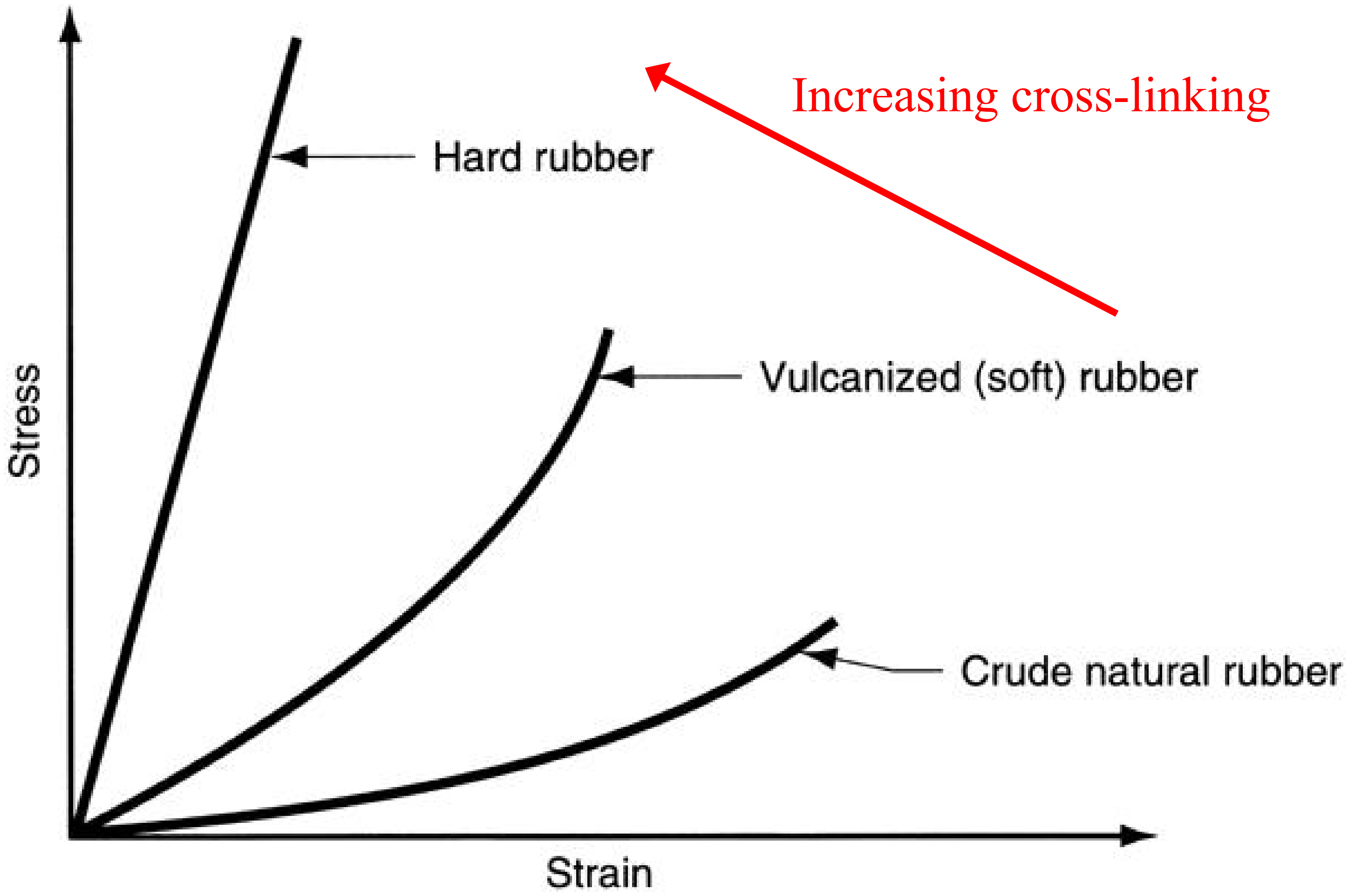


Hard and strong



Hard and tough

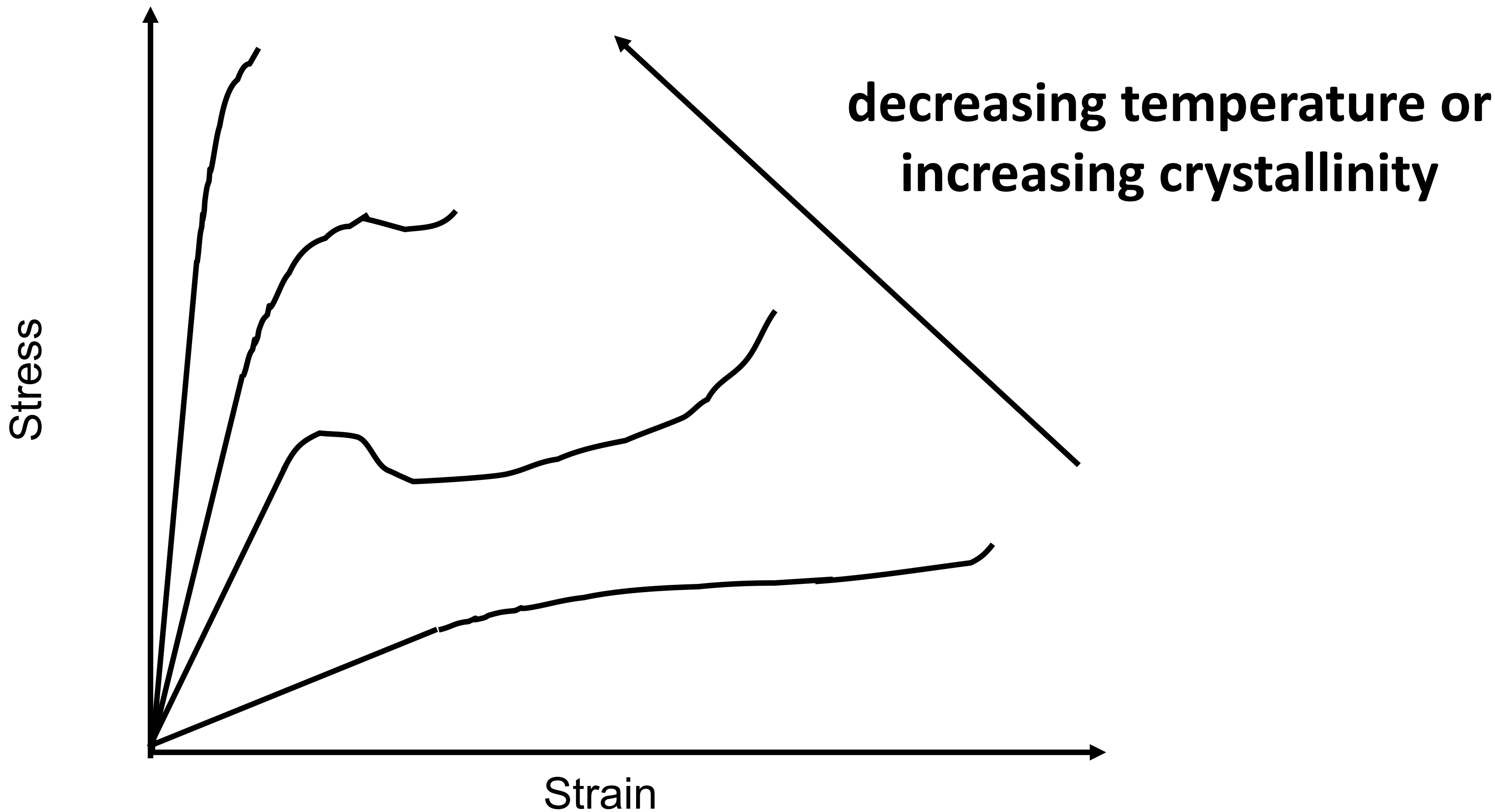
Properties depend on amount of cross-linking

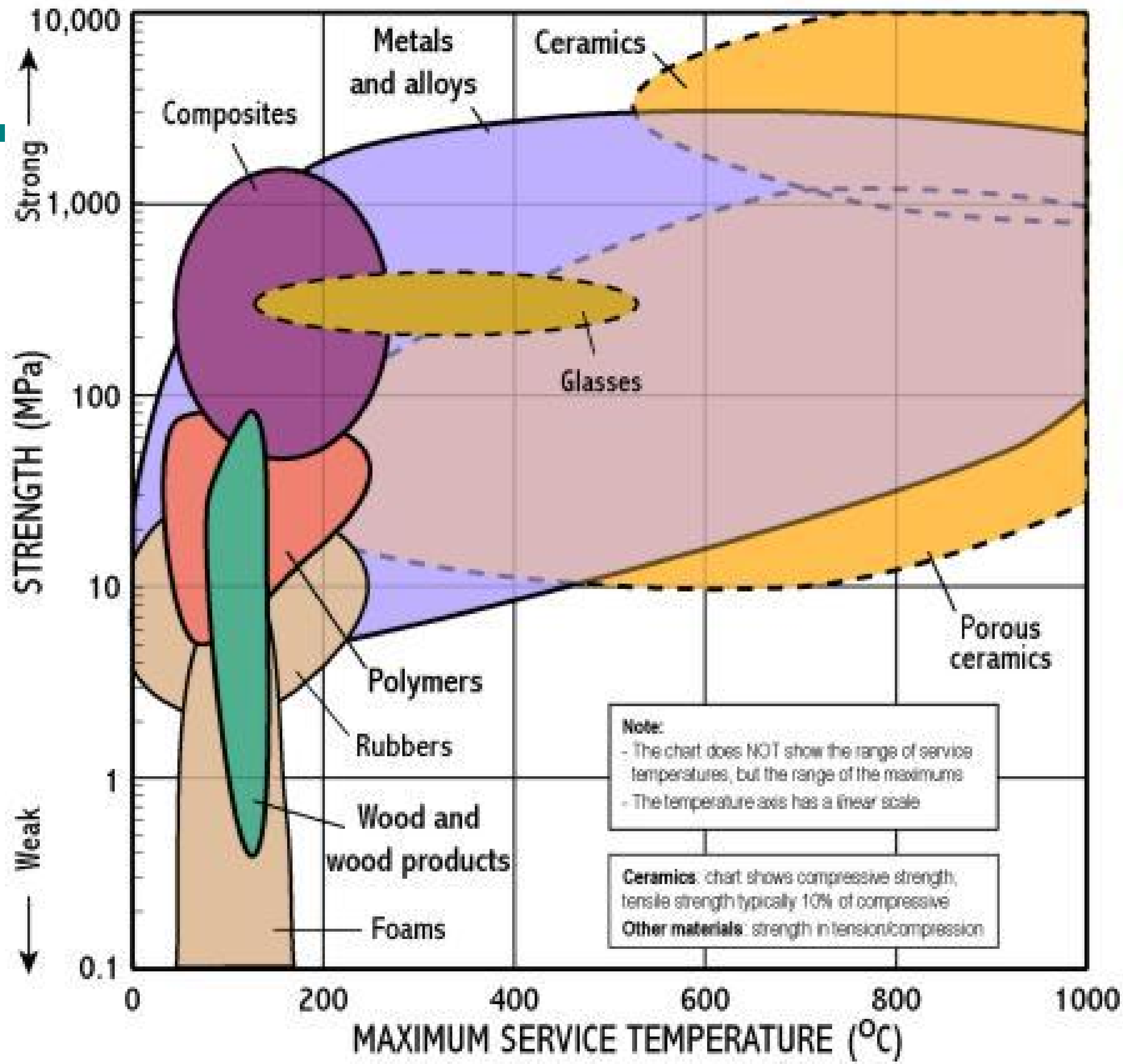


Polymers: Thermal Properties

- In the liquid/melt state enough thermal energy for random motion (Brownian motion) of chains
- Motions decrease as the melt is cooled
- Motion ceases at “glass transition temperature”
- Polymer hard and glassy below T_g, rubbery above T_g

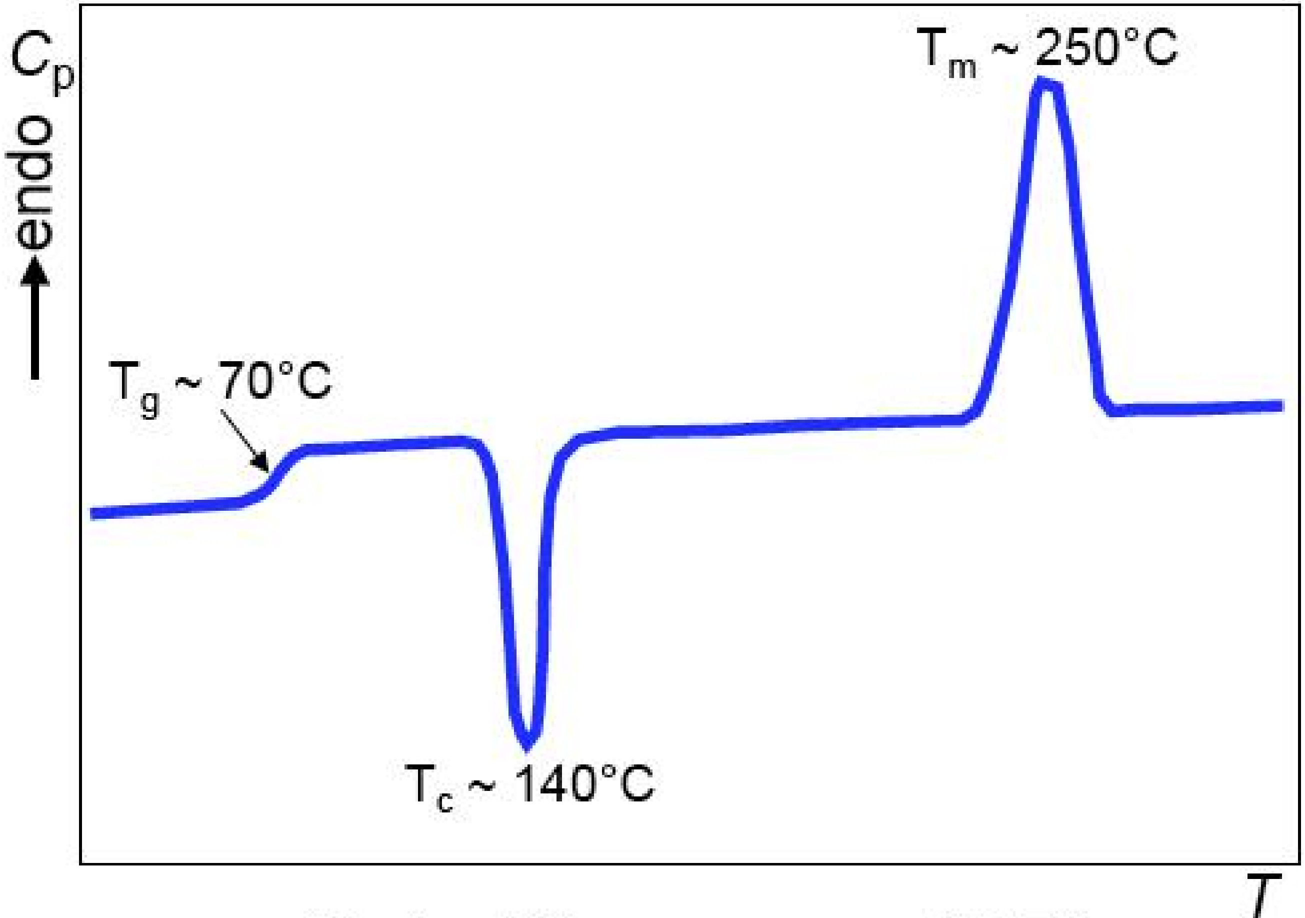
Thermal Properties





10,000
1,000
100
10
1
0.1
Strong
Weak
STRENGTH (MPa)
0
200
400
600
800
1000
MAXIMUM SERVICE TEMPERATURE (°C)
Metals and alloys
Ceramics
Composites
Glasses
Polymers
Rubbers
Wood and wood products
Foams
Porous ceramics
Note:
- The chart does NOT show the range of service temperatures, but the range of the maximums
- The temperature axis has a linear scale
Ceramics: chart shows compressive strength, tensile strength typically 10% of compressive
Other materials: strength in tension/compression

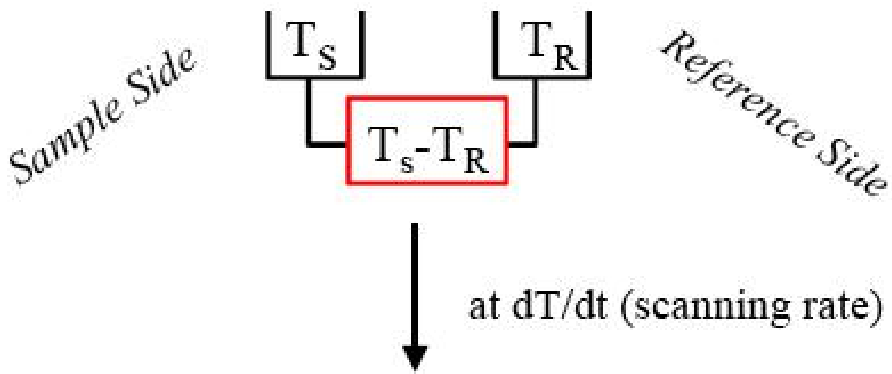
1. Crystallization and melting of crystalline polymers.
2. Glass transition of amorphous polymers.

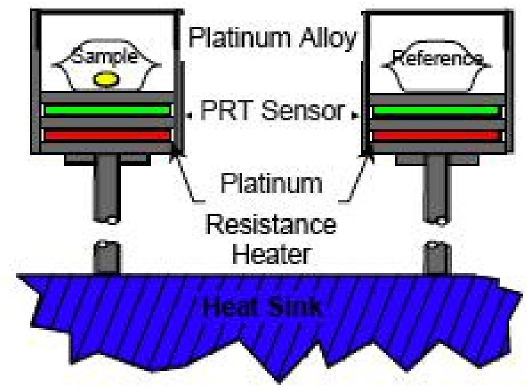


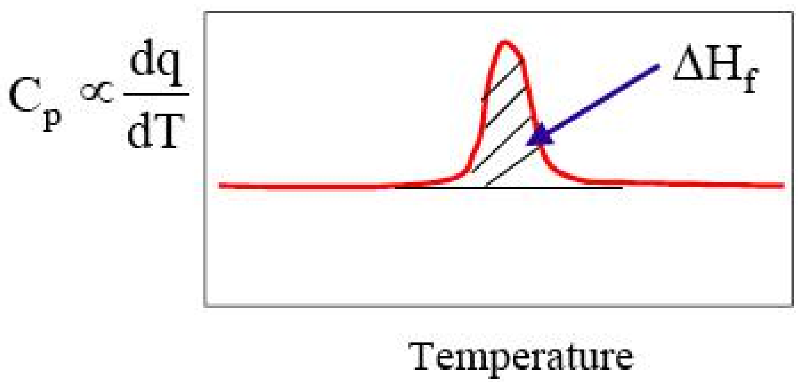
Typical Thermogram of PET

Differential Scanning Calorimetry (DSC)

A thermal analysis technique suited for observing the melting behavior of polymers.







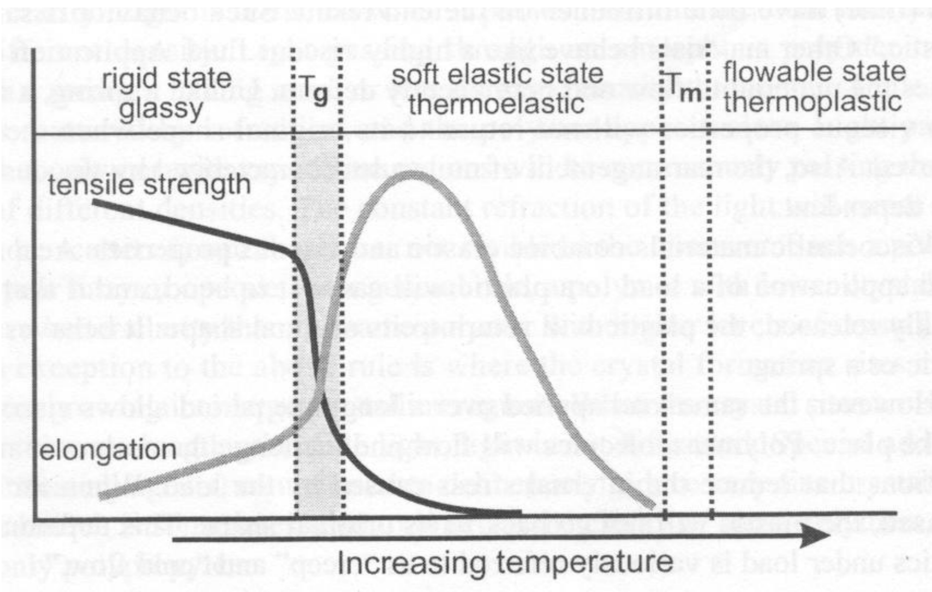
$$\boxed{w^c = \frac{\Delta H_f}{\Delta H_f^0}}$$

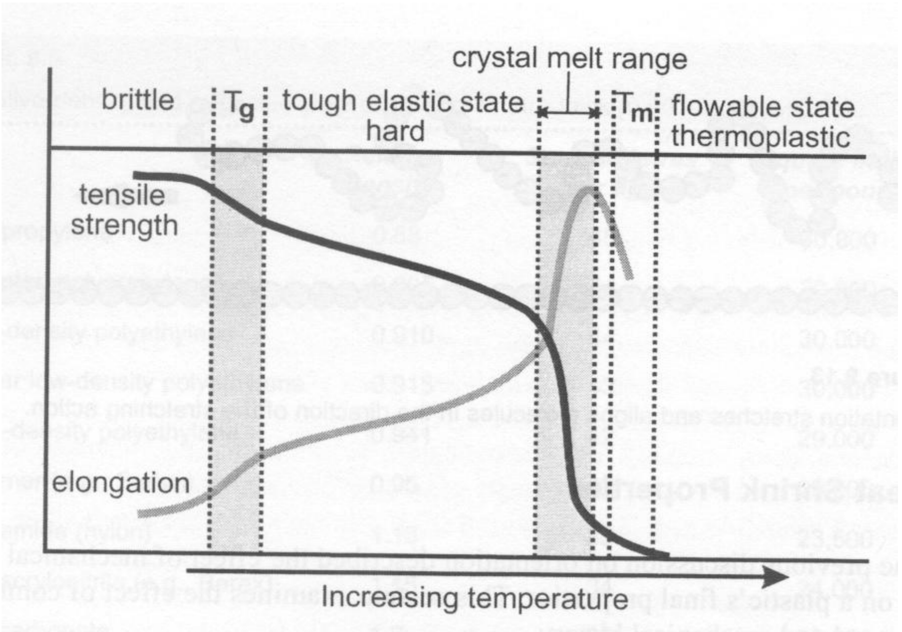
heat of fusion for perfect (100%) crystals

The observed melting behavior of polymers is dependant on the previous thermal history due to its role in crystallite size and total degree of crystallinity.

Thermo-mechanical Properties

Amorphous v Crystalline Polymers



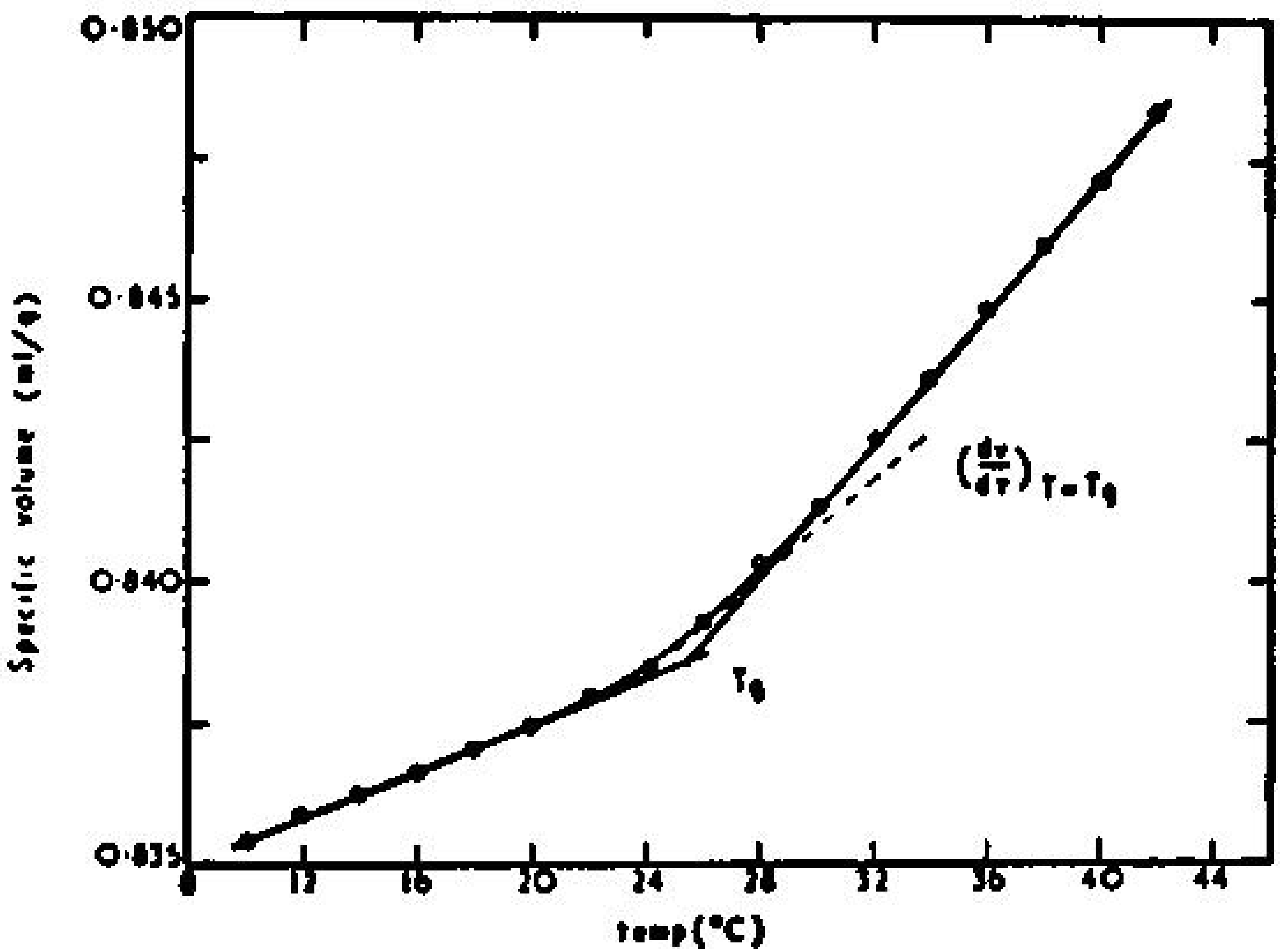


Thermal Expansion

- **If a part is to be produced within a close dimensional tolerance, careful consideration of thermal expansion/contraction must be made.**

- **Parts are produced in the melt state, and solidify to amorphous or semi-crystalline states. Changes in density must be taken into account when designing the mold.**

Thermal Expansion



Thermal Expansion

Metals	*Coefficient of Expansion*
Aluminum	23.5
Brass	18.8
Copper	16.7
Steel	10.8
Plastics	
Poly(vinylidene chloride)	190 - 200
Polyethylene	110 - 250
Polystyrene	60 - 80
Polyamide (nylon)	90 - 108
Phenolic (thermoset)	30 - 45

Heat Distortion Temperature

- The maximum temperature at which a polymer can be used in rigid material applications is called the softening or heat distortion temperature (HDT).

- A typical test (plastic sheeting) involves application of a static load, and heating at a rate of 2^oC per min.

- The HDT is defined as the temperature at which the elongation becomes 2%.

Heat Distortion Temperature

A: Rigid poly(vinyl chloride)

50 psi load.

B: Low-density poly(ethylene)

50 psi load.

C: Poly(styrene-*co*-acrylonitrile)

25 psi load.

D: Cellulose acetate

(Plasticized) 25 psi load.